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16 September 1977
SUPERSEDING
MIL-C-43214C
28 March 1973

MILITARY SPECIFICATION

CAMERAS, COPYING, LITHOGRAPHIC PROCESS, DARKROOM TYPE

This specification is approved for use by Natick Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type and three sizes of lithographic process copying cameras (see 6.1).

* 1.2 Classification. The cameras covered by this specification shall be of the following sizes as specified (see 6.2).

Size 1 - Deleted (see 6.6)
Size 2 - 20 x 24 inch (508 mm x 610 mm) film
Size 3 - 24 x 24 inch (610 x 610 mm) film
Size 4 - Deleted (see 6.6)
Size 5 - 31 x 31 inch (762 x 762 mm) film

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

* CC-M-1807 - Motors, Alternating Current, Fractional and Integral Horsepower (500 Horsepower and Smaller)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3610

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DD-G-451 - Glass, Plate, Sheet, Figured (Float, Flat, For Glazing, Corrugated Mirrors and Other Uses)

MILITARY

MIL-G-174 - Glass, Optical
MIL-C-675 - Coating of Glass Optical Elements (Anti-Reflection)
MIL-C-3615 - Cameras, Copying Lithographic, Packaging Of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-150 - Photographic Lenses
MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements For Equipment
MIL-STD-462 - Electromagnetic Interference Characteristics Measurement Of

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL BUREAU OF STANDARDS

Handbook H-28 - Screw Thread Standards For Federal Services

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

Std. No. 70 - The National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups, and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a first article for inspection and approval (see 4.3, 6.2 and 6.3).

3.2 Electrical code. All electrical wiring shall conform to the NFPA Standard No. 70.

3.2.1 Certification. Prior to the approval of either the preproduction sample or initial production unit or if none is submitted, prior to the approval of the first shipment, the contractor shall submit acceptable evidence that the equipment he proposes to supply under this specification meets the requirements of the NFPA Standard No. 70. Acceptable evidence of meeting these requirements shall be a certified test report from a nationally recognized independent testing laboratory approved by the contracting officer stating that the electrical equipment offered has been tested and conforms with the NFPA Standard No. 70.

3.3 Materials. Materials shall be as specified herein. Materials not definitely specified shall be the same as normally used by the manufacturer for darkroom cameras provided the completed items comply with all provisions of this specification. Excluding the bellows, darkroom seal material, rubber, plastic, and glass components, the camera shall be constructed of metals. Use of lightweight, high strength metal alloy shall be utilized wherever practicable to keep the weight of the camera at a minimum. Wood or wood products shall not be used as part of the camera. Materials subject to corrosion shall be either corrosion-resisting or treated to resist corrosion (see 6.5).

3.3.1 Plate glass. Plate glass shall conform to type I, class 1, quality q1 or q2 of DD-G-451. The thickness of the plate glass shall be 1/4 inch (6.4 mm) nominal.

3.3.2 Optical glass. Optical glass for the lens shall conform to class 1 or 2, grade A, plate or slab of MIL-G-174.

3.4 Design and construction. The cameras shall be a darkroom precision type. The design and construction of the cameras shall provide positive locking of all movable components. The cameras shall be of a horizontal stand design with a stationary back. The movement of components controlling the operation, focusing, and adjustment of the cameras shall be positive with uniform resistance, and freedom from lost motion. Controls shall be located within reach of the operator and in such a manner as not to interfere with the operation of the camera.

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3.4.1 Van requirement, Army. When specified for Army van use (see 6.2), the size 5 camera shall be capable of withstanding, without harm, the stress, vibration and shock resulting from movement in a van across rough roads as specified in 4.6.12.

3.4.2 Capacity. The camera back shall have a capacity to accommodate sheet film and a contact screen in the sizes shown in table I, for line and half-tone reproduction. Each camera shall be supplied with a magenta (positive or negative) or gray (positive or negative) type contact screen with 120, 133 or 150 ruling line pairs per inch (6 ruling line pairs per mm), as specified (see 6.2).

TABLE I. Film and screen sizes, nominal inches (mm)

Size (camera)	Film, in. (mm) Minimum	Maximum	Contact screen, in. (mm)
2	4 x 5 (102 x 127)	20 x 24 (508 x 610)	21 x 25 (533 x 635)
3	4 x 5 (102 x 127)	24 x 24 (610 x 610)	25 x 25 (635 x 635)
5	4 x 5 (102 x 127)	31 x 31 (787 x 787)	32 x 32 (813 x 813)

3.4.3 Dimensions and weight. The overall dimensions and weight of the cameras shall not exceed the values shown in table II.

TABLE II. Dimensions and weight

Size of camera	Length in. (m)	Width in. (m)	Height in. (m)	Weight net lbs. (kg)
2	204 (5.2 m)	48 (1.2 m)	80 (2 m)	2500 (1130 kg)
3	210 (5.3 m)	53 (1.4 m)	85 (2.2 m)	2750 (1250 kg)
5	175 (4.4 m)	65 (1.7 m)	85 (2.2 m)	3500 (1590 kg)

3.4.4 Range of reproduction. The minimum reproduction range of the cameras shall be as shown in table III when equipped with the lenses specified in 3.4.5.4.1.

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TABLE III. Range of reproduction

Size of camera	Reduction (percent)	Enlargement (percent)
2	20	200
3	20	200
5	25	300

3.4.5 Detail of components.

3.4.5.1 Copyboard assembly. The copyboard assembly for the cameras shall consist of a carriage and a hinged, counterbalanced, metal framed, glass covered, rubber or plastic foam padded, spring loaded copyboard. The copyboard shall accommodate copies of 1/2-inch (13 mm) maximum thickness in the sizes shown in table IV. The cover glass shall be maintained in parallel alignment with the camera focal plane when the board is in the vertical position. The glass shall be as specified in 3.3.1. Graduated centerlines shall be permanently marked on the copyboard pad to provide a guide for centering the copy. The copyboard frame shall be pivot mounted on the carriage. The pivot mounting shall have spring-loaded locking means to permit the copyboard to be secured in the vertical position for copying and in the horizontal position for loading. The copyboard shall have its center coincident with the optical axis of the lens. The assembly shall be mounted on the base assembly specified in 3.4.5.2, opposite the end containing the camera back assembly. The copyboard carriage shall be mounted on roller bearings or roller bushings on the track of the base assembly, opposite the end with the camera back assembly and shall maintain the copyboard of the size 2 and 3 cameras in parallel alignment within 0.002-inch (.051 mm) tolerance, in all positions of operation between copyboard, lensboard and vacuum back and ground glass assemblies, when tested as specified in 4.4.2.3. The carriage shall have a positive locking device for holding the copyboard assembly at any position along the track. The copyboard carriage drive shall be manual except when specified to be automatic motor driven (see 6.2). Size 5 cameras shall be capable of maintaining their parallel alignment within .005-inch (.127 mm) tolerance when tested as specified in 4.4.2.3 when supporting copy weighing up to 25 pounds (11.3 kg). Alignment tests for size 5 cameras shall be conducted after vibration resistance tests are completed.

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TABLE IV. Copyboard capacity

Size of camera	Size of copy inches maximum (mm)
2	30 by 40 (762 x 1016)
3	34 by 44 (864 x 1118)
5	40 by 50 (1016 x 1270)

3.4.5.2 Base assembly. The base assembly of the camera shall include a smooth rail or ways mounted in rubber mounts or other shock absorbing devices that shall absorb shock within 10 seconds when tested as specified in 4.6.1. The rail or ways shall be sufficiently sturdy to support and maintain complete parallel alignment in all positions of operation between the copyboard, lensboard and camera back assemblies, when tested as specified in 4.6.2. The rail or ways shall be attached to the back assembly floor mount and support for one end of the rail or ways and to a second floor mount and support for the opposite end. The floor mounts shall have bolt holes for attachment of the camera to a floor.

3.4.5.3 Lamp carriers. The lamp carriers shall consist of vertical lamp supports and adjustable horizontal arms. The carrier arms shall accommodate lamps and be capable of being swung horizontally about a pivot point to permit the lamp arms to be placed in all positions necessary for uniform illumination of the copyboard and hand locked. The lock shall operate positively with no binding, when tested as specified in 4.6.3.

3.4.5.4 Lamps. The lamps for copyboard illumination shall be pulsed xenon arc, quartz iodide, mercury vapor. Each lamp shall have a reflector. The lamps shall be accommodated in the carriers specified in 3.4.5.3 to provide uniform illumination of the copyboard within a tolerance of plus or minus 15 percent, with no blistering or discoloration of the reflectors, when tested as specified in 4.6.3. A heavy duty power pack built for continuous use with light controls shall be provided for control of the light intensity of the lamp.

3.4.5.5 Lensboard assembly. The lensboard assembly shall consist of a movable carriage, lensboard, lens, flash lamp, manual iris diaphragm control and shutter with electric timer. The lens case assembly shall be of sufficient size and strength to hold all components in proper alignment. The carriage shall maintain the lens case in parallel alignment in all positions of operation between the copyboard, lensboard and camera back assemblies, when tested as specified in 4.6.2.

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3.4.5.5.1 Lens. The lens as specified in 3.4.2 shall have surfaces free of scratches or stains. The radial distortion of the lens shall not vary by more than 0.05 millimeter (mm) at a ratio of 1:1 reproduction, throughout an angular field of 0 to 23 degrees, and the lens shall have characteristics as shown in table V, when tested as specified in 4.6.4. All glass-air surfaces shall be given a reflection reduction coating conforming to MIL-C-675. The lens shall be calibrated by the contractor in accordance with table V and shall have a matching cap. The lens shall be capable of being mounted on the lens plate and attached to the lens board. A metal box attached to the base assembly shall be provided for storage of the lens. When specified (see 6.2), two interchangeable, color corrected, process lenses conforming to MIL-STD-150, type IV, and as shown in table V, shall be provided with each size 5 camera. Unless otherwise specified (see 6.2), 24-inch (610 mm) lens will be furnished.

TABLE V. Lens characteristics

Size	Equivalent focal length in. (mm)	Max. relative aperture	Min. relative aperture	Field of view (full angle) (degrees)	Color correction
2 and 3	19 (483)	f/11	f/64	42 - 46	apochromatic
5	19 (483) and 24 (610)	f/11	f/64	42 - 46	apochromatic

3.4.5.5.2 Iris diaphragm. The iris diaphragm shall have a manual control for correlation of the "f" values of the lens to provide an equivalent transmission of light at various ratios of enlargement and reductions. The diaphragm control shall be marked as follows: f/11, f/16, f/22, f/32, f/45 and f/64 with intermediate graduations.

3.4.5.5.3 Flash lamp. The lensboard assembly shall be provided with a spring-loaded, counterbalanced, pivoting flash lamp. The lamp shall be capable of being moved in front of the camera lens, and being moved out of range of the optical system when the lamp is not in use. The lamp shall use a diffused light source containing a 60 watt bulb. The lamp shall be controlled from the gallery.

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3.4.5.5.4 Shutter. The shutter shall operate by a solenoid controlled from the darkroom or gallery. The solenoid shall open and close the shutter by manual operation of a switch or automatic operation of an electric timer preset to a predetermined time interval. The timer shall be calibrated in minute and second intervals and be capable of recycling or of being reset for different cycles, when tested as specified in 4.6.5.

3.4.5.6 Bellows assembly. A bellows assembly shall connect the lensboard assembly to the camera back assembly. The complete bellows assembly shall be readily removable for ease of maintenance and periodic calibration of the camera. The bellows shall provide the necessary extension and retraction to obtain reproduction within the range specified in 3.4.4 without image obstruction in any position. The bellows shall be fabricated of molded rubber or impregnated rubberized materials, and shall be an accordian type with all corners reinforced and supported along its length to prevent sag. The bellows frame shall fit properly to lensboard and vacuum back. The bellows shall permit adjustment movements with no damage or sag, and shall be light-tight when tested as specified in 4.6.6.

3.4.5.7 Back assembly. The camera back assembly shall include a vacuum back and the ground viewing glass, separately mounted to permit the vacuum back and glass to be alternately brought into the focal plane of the camera. The back assembly shall be attached to the base assembly and be in parallel alignment in all positions between the camera back, lensboard, and copyboard assemblies (see 4.6.2).

3.4.5.7.1 Vacuum back. The vacuum back shall be hinged to pivot about a vertical or horizontal axis. The vertical dimension of the back shall be the larger dimension. The vacuum surface shall be black, neutral, or of low-light reflecting color and have small perforations or slots for holding film and contact screens by vacuum. A mechanical means shall be provided for applying vacuum to the areas required for applicable film and screen sizes. The vacuum back shall hold 0.003 thru 0.007-inch (.076 thru 178 mm) thick film smoothly, emulsion side up, with no visual distortion of the surface, when tested as specified in 4.6.7.

3.4.5.7.2 Ground viewing glass. The ground viewing glass shall conform to 3.4.1. The glass shall be housed in a metal frame hinged to pivot about a vertical or horizontal axis so as to permit the glass to be swung away from the viewing position when the glass is not in use. One surface of the glass shall be uniformly fine grained obtained by an acid or mechanical process. The glass shall be free from scratches and stains and capable of resolving not less than 20 lines per millimeter in a projected image, when tested as specified in 4.6.8. The glass shall have permanent markings that shall be coincident

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with markings on copyboard when tested as specified in 4.6.8. The glass permanent markings shall also permit visual orientation and positioning of the image in relation to the film position on the vacuum back. Orientation markings shall consist of a clear, ungrained glass of 1-inch (25.4 mm) square or circle at the center and diagonals radiating to the corners of the glass. The edges of the glass shall be ground square.

3.4.5.7.3 Vacuum system. The camera vacuum back shall be equipped with a motor-driven vacuum pump and tubing. The pump shall be either a multi-stage turbine, or an oilless, graphited-vane, rotary pump of adequate vacuum capacity to hold the sizes of film as specified in 3.4.2. The vacuum pump shall be furnished complete with base, switches, controls, fittings, vacuum gage, exhaust muffler, vibration mounts, and connecting hose. The motor, specified in 3.4.7 shall drive the pump. The motor and pump shall be mounted and aligned as a unit on a base assembly. A switch shall be provided for control of the motor. The vacuum system shall be capable of producing a 24-inch (610 mm) (mercury) vacuum within 1 minute, when tested as specified in 4.6.9.

3.4.5.8 Darkroom wall-opening adapter. The cameras shall have a complete darkroom wall-opening adapter. The adapter shall be flexible and provide a light-tight joint between the camera back and the darkroom wall opening when tested as specified in 4.6.6. The adapter shall be black and fabricated from an impregnated rubber material.

3.4.5.9 Focusing controls. Controls for focusing shall include glass indicator(s) with a hairline mark on both sides of the glass and two metal or plastic focusing tapes or counting devices that shall be legible and related to the lens. The focus measuring tapes or counting device shall measure distances of lens to copy and lens to film to obtain the reproduction ratio required. Unless otherwise specified (see 6.2), hand controls for movement of the lensboard and copyboard carriages for focusing shall be located on the camera back assembly. A computer of the linear or radial slide rule type shall be furnished, for converting reproduction ratios into the correct readings for setting on the focusing scales or counting device. A direct reading scaling system using focusing tapes for copyboard and lensboard in direct reading percentages of original size may be used in lieu of the above measuring system and computer. With the lensboard and copyboard set for a 1 to 1 reproduction ratio the camera shall take a sharp picture of the test grid or two test scales, as applicable (see 3.4.6) that shall have a reproduction accuracy within 1/64 inch (.4 mm) in the vertical and horizontal directions throughout the entire field of view, when tested as specified in 4.6.10.

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3.4.6 Reproduction test aid. A reproduction test aid consisting of a rectalinear grid equal in area size to the copyboard or two 18-inch (457 mm) long calibrated scales shall be furnished with each camera, for testing and checking the reproduction ratio accuracy. The grid shall consist of 0.250 inch (6 mm) squares and the scale shall consist of 0.250 inch (6 mm) graduations with a tolerance of plus or minus 0.005 inch (.127 mm) for each. A protective container shall be provided for storage of the test grid or test scales, as applicable.

3.4.7 Electrical components and motors. The electrical components shall operate on 120/220 volts, 60 hertz, single phase alternating current. All wiring shall conform to NFPA Standard No. 70. The wiring shall be enclosed in metallic tubing, where possible. The cords extending to the carriages shall be rubber covered with helical coils, or be provided with automatic take ups. All wiring shall be color coded. Terminal boxes shall be provided for all connections. A plastic inclosed or an etched metal wiring diagram, detailing color coded wires, and identifying switches and controls by name and part number, shall be mounted on a terminal box. The motors shall conform to CC-M-1807, continuous duty, 1/4 to 1/2 horsepower (187 to 373W) as applicable. When specified (see 6.2), motors shall be fungus proofed in accordance with applicable requirements of CC-M-1807.

* 3.5 Contract data requirements. The contract data requirements shall be as specified (see 6.2 and 6.2.1).

3.6 Electromagnetic compatibility. When specified (see 6.2), the camera shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461 (see 4.6.11).

3.7 Welding. All welds shall be adherent, and free of cracks. Spot or tack welds in aluminum shall be limited to lightly stressed, noncritical structures. Welding shall be cleaned and free from scale, or flux deposit. Acid used for cleaning shall be completely neutralized and removed. Welds shall be free from spatter, burn through, blisters, and pin holes.

3.8 Screw threads. All screw threads shall conform to the National Bureau of Standards Handbook H-28.

3.9 Lubrication. Oil fittings, grease cups, or other means of lubrication shall be provided for lubricating all parts subject to wear. Parts that must not be lubricated, such as graphited vanes in the vacuum pump and the lens iris diaphragm, shall be marked with a "DO NOT LUBRICATE" warning.

3.10 Finish. The cameras shall be given the finish normally used by the manufacturer on this type of equipment.

3.11 Marking. Marking for identification shall be in accordance with manufacturer's standard commercial practice and shall include manufacturer's name, date of manufacture, and model number.

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3.12 Instruction plates. Instruction plates shall be provided legibly describing any special or important procedures to be followed during operation or servicing, including warnings and cautions.

3.13 Tools. Each camera shall be provided with tools necessary for operation.

3.14 Workmanship. The finished camera shall be free from burrs, slivers or sharp edges, dents, fractures, breakage, and misalignment of parts. Bolts, nuts, and screws shall not be missing, loose, fractured, broken, or stripped.

3.15 Government loaned property. The Government will furnish one tractor and van (truck, camera, map reproduction) for the first article test, when applicable, for size 5 cameras.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

* 4.3 First article. When a first article is required, the camera shall be examined for defects listed in table VI, dimensions specified, weight specified, and tested in accordance with 4.6 thru 4.6.10 and, when applicable, 4.6.11 and 4.6.12. The presence of any defect, any dimension or weight not within the specified limits, and the failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

* 4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

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* 4.4.2 End item inspection. The inspection lot shall consist of all complete cameras of one size offered for inspection at one time. The sample unit shall be one complete camera.

* 4.4.2.1 Visual examination. The cameras shall be examined for the defects in table VI. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE VI. Classification of defects

Examine	Defect	Classification Major	Minor
Design and construction	Locking of moving components not positive	X	
	Movement of operating components for focusing and adjustment not positive with uniform resistance and free from lost motion	X	
	Controls not within reach of operator, or interfere with operation of camera	X	
Contact screen	Missing or not as specified		X
Copyboard	Pad missing, glass loose, not framed in metal, scratched, cracked, centerlines missing, not permanently marked, not pivot mounted, not counter-balanced, locking device missing or carriage not as specified	X	
Base	Rail or ways not smooth	X	
	Shock absorbers missing or not as specified	X	
	Lens storage box missing	X	
Lamp carriers	Not as specified	X	
	Lamps missing or do not operate as specified	X	

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TABLE VI. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Lens and lensboard	Not as specified including lens not free from scratches or stains; cap, reflection coating, or calibration missing	X	
Iris diaphragm	Not graduated as specified or graduations not legible	X	
Flash lamp	Not as specified or broken	X	
	Bulb missing or does not operate		X
Shutter	Shutter does not open or close by solenoid when solenoid is energized	X	
Bellows assembly	Material not as specified, cracked or worn, corners not reinforced or not supported to prevent sag	X	
	Improper bellows frame fit to lensboard or vacuum back	X	
Back assembly	Not as specified, including marking missing or not legible, ground viewing glass, markings not as specified, missing or not legible, or vacuum system cannot be brought into focal plane of camera	X	
Darkroom wall-opening adapter	Not as specified	X	
	Worn, cracked or torn	X	
Focusing controls, including aids	Missing or not as specified	X	
Ground glass	Not free from scratches or stains	X	
Electrical components	Not as specified	X	

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TABLE VI. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Welding	Not adherent or free from cracks	X	
	Spot welds not as specified	X	
	Not clean and free from scale or flux deposits		X
	Burn through, spatter, blisters, or pin holes		X
* Bolts, nuts and screws and other types of threaded components	Loose or missing, fractured, broken, or stripped	X	
Lubrication fittings	Missing	X	
Finish	Not adherent and free from runs, tackiness, embedded foreign matter, or areas of no finish		X
Marking for identification	Not as specified or missing	X	
Instruction plates	Missing, or not legible		X
Tools	Operational tools missing		X
Workmanship	Not free from burrs, slivers or sharp edges		X
	Not free of dents, fractures, breakage, and misalignment of components	X	

* 4.4.2.2 Dimensional examination. The cameras shall be examined for compliance with dimensions and weight specified. Any dimension or weight not within this specification tolerance shall be a defect. The inspection level shall be S-2 with an AQL of 4.0 defects expressed in terms of defects per hundred units.

* 4.4.2.3 End item testing. Each camera shall be as specified in 4.6 thru 4.6.10 and 4.6.12. Failure of any test shall be cause for rejection of the item. When applicable, the test specified in 4.6.11 shall be performed on one representative production unit (see 6.7). Failure of the test shall be cause for rejection of the lot.

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4.5 Certificate examination. Certificate of compliance, or certified test reports, as applicable, that are submitted as proof of conformance with specification requirements shall be examined and validated.

4.6 Test methods. Tests shall be conducted at an ambient temperature of between 68 to 75F (20 to 24C) and at 45 to 60 percent relative humidity.

4.6.1 Shock test. A 50-pound (22.7 kg) weight in free-fall shall be dropped from a 3-foot (914 mm) height on a concrete floor at 10 feet (3.1 m) from the copyboard end of the camera. With a test target consisting of a calibrated focusing screen mounted on the copyboard and the camera set for a 1:1 reproduction ratio, the image of the target shall be observed on the ground viewing glass for detection of vibration or other movements resulting from the weight drop. Any nonconformance with the 10-second shock recovery requirements of 3.4.5.2 shall constitute failure of this test.

4.6.2 Parallelism test. The parallelism test of copyboard, lensboard and camera back assemblies shall be made by means of a telescoping trammel rod with dial indicator (calibrated in 0.001-inch (.025 mm) increments) attached at one end. Parallelism of the planes of these components shall be checked at reproduction ratios of 50, 33-1/3, 25, and 20 percent reduction and 200 percent enlargement. Any nonconformance with the parallel alignment in all positions between copyboard, lensboard, and camera back requirements of 3.4.5.2, 3.4.5.5 and 3.4.5.7 shall constitute failure of this test.

4.6.3 Lamp carrier binding and illumination tests. A sheet of white paper equal in size to the copyboard and equally subdivided into 25 rectangles shall be placed on the copyboard. The lamp carriers shall be positioned as far as possible away from the copyboard. The lamps shall be turned "on" and adjusted to obtain maximum illumination coverage of the copyboard and lamp carriers locked. All other lights shall be turned "off". Light meter readings at the center of each of the 25 rectangles shall be taken and the reflecting surfaces of each of the lamps shall then be examined. Any nonconformance with lamp carrier locking requirements of 3.4.5.3 and the uniform illumination and reflector blistering and discoloration requirements of 3.4.5.4 shall constitute failure of this test.

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4.6.4 Lens test. The contractor shall arrange to have the lens tested for characteristics specified in table V in accordance with test methods in MIL-STD-150 except for the apochromatic characteristic. A certified certificate from the lens manufacturer stating that the lens is an apochromatic lens is acceptable. Any nonconformance with the other lens characteristic requirements of 3.4.5.5.1 shall constitute failure of this test.

4.6.5 Shutter test. The timer on the shutter shall be set and tested for 5-, 15-, and 50-second and 2-, 3-, and 5-minute intervals. Any nonconformance with the timer recycling and resetting requirements of 3.4.5.5.4 shall constitute failure of this test.

4.6.6 Light-tightness test. The camera shall be set up with a darkroom. The lens shutter shall be "closed" and the lensboard assembly shall be moved forward toward the copyboard until the bellows extends to 1/2 its maximum limit. With copyboard illumination lamps directed toward the lensboard and bellows and turned on, all room lights "off", and a 5-minute allowance for eyes to become accustomed to darkness, the bellows shall be extended and retracted to its maximum limit for a minimum of 5 cycles. The bellows at its maximum extension limit shall then be examined for light-tightness and bellows damage and sag. Any nonconformance with the bellows sag and light-tight requirements of 3.4.5.6 and the darkroom wall-opening adapter light-tight requirements of 3.4.5.8 shall constitute failure of this test.

4.6.7 Vacuum back test. The vacuum back shall be tested for film adherence with 6 films, ranging from 0.003 to 0.007-inch (.076 to .178 mm) thick and ranging in sizes from 4 by 5 inches (102 by 127 mm) to the maximum film size that the back accommodates. Each film shall be tested separately. A light shall then be moved about the film surface to check distortion. Any nonconformance with the visual distortion of film surfaces of 3.4.5.7.1 shall constitute failure of this test.

4.6.8 Ground viewing glass test. The ground viewing glass shall be tested for resolution of lines per millimeter and the coincidence of ground glass and copyboard markings. Any nonconformance with the line resolution and coincidence of ground viewing glass and copyboard marking requirements of 3.4.5.7.2 shall constitute failure of this test.

4.6.9 Vacuum pump test. The pump shall be tested to determine the length of time required for the pump to produce a 24-inch (.6 m) (mercury) vacuum. Any nonconformance with the vacuum requirements of 3.4.5.7.3 shall constitute failure of this test.

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4.6.10 Focusing test. The reproduction test grid or two test scales (see 3.4.6), as applicable, shall be placed on the copyboard. The two test scales shall be centrally located at right angles to each other. The lensboard and copyboard shall be set for a 1 to 1 reproduction ratio and the lens operation at f/22. The copyboard lamps shall be placed 2 feet (610 mm) from the center of the copyboard at an angle of 45 degrees and the light source shall be set for 3200 kelvin. With the shutter control on "manual" all room lighting shall be turned "off" and the lamps turned "on". The picture of the reproduction test grid or scales shall be photographed for determination of picture sharpness and accuracy of reproduction. Any nonconformance with the reproduction accuracy requirements of 3.4.5.9 shall constitute failure of this test.

4.6.11 Electromagnetic compatibility test. When electromagnetic compatibility is required, the first article or initial production unit as applicable, shall be tested by the contractor in accordance with test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the contractor or an independent testing agency. The contractor shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan and the Electromagnetic Test Report and the requirements meet MIL-STD-461.

* 4.6.12 Vibration resistance test (size 5 camera). When specified (see 3.4.1 and 6.2) for Army van use, the size 5 camera shall be mounted in the Government loaned van and shall be positioned for transit. The complete vehicle with the camera mounted therein shall be driven a minimum of 20 miles (32 km). Not less than 10 miles (16 km) shall be unsurfaced, stabilized, or partially surfaced earth or gravel roads. The test shall include ten abrupt starts, ten abrupt stops, and five sharp turns at maximum safe speeds in each direction (right and left). The route shall be approved by the contracting officer. At the completion of the test run, the camera shall be operated. Any evidence of damage to the camera or inability of the camera to reproduce within its rated capacity of enlargement and reduction and within its field of lens coverage, all copy in accurate size and proportion to within 0.005 inch (.127 mm) of the required dimensions shall constitute failure of this test.

* 4.7 Inspection of packaging. An inspection shall be made to determine that preservation-packaging, packing and marking as specified in section 5 is in conformance with MIL-C-3615.

5. PACKAGING

* 5.1 Preservation-packaging, packing and marking. Preservation-packaging, packing and marking for level A, B, or C as specified (see 6.2), shall be in accordance with MIL-C-3615.

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6. NOTES

6.1 Intended use. The cameras covered by this specification are horizontal and intended for use in the precision production of line and half tone negatives for the lithographic process. Size 5 cameras when specified (see 6.2), for use in mobile vans are intended for map reproduction with Army Topographic Support System.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size of camera required (see 1.2).
(c) When a first article is required (see 3.1, 4.3 and 6.3).
(d) When camera is for Army van use (see 3.4.1).
(e) When electromagnetic compatibility is required (see 3.6).
(f) Type of contact screen with ruling line pairs per inch required (see 3.4.2).
(g) When automatic copyboard drive is required (see 3.4.5.1).
(h) When two lenses are required for size 5 camera (see 3.4.5.5.1).
(i) When other than 24 inch (610 mm) lens is required (see 3.4.5.5.1).
(j) When focusing controls are to be located other than on the camera back assembly (see 3.4.5.9).
(k) When fungus proofing of motor is required (see 3.4.7).
(l) Contract data (see 3.5).
(m) Selection of applicable levels of preservation-packaging and packing (see 5.1 and 5.2).

* 6.2.1 Contract data requirement. Any data items to be delivered under any contract covered by this specification shall be specifically called for in contract in accordance with applicable regulation of the procurement activity (see 3.5 and 6.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 Government loaned property. The contracting officer should arrange to loan the property specified in 3.14.

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* 6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

* 6.6 Deletion data. Sizes 1 and 4 of MIL-C-43214C were deleted from this revision. Size 1 is not required. Size 4 is not available and shall be replaced by size 5.

6.7 Changes from previous issue. The outside margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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